

2022 Ph H2 Q6

A distant galaxy emits light at a wavelength of 486 nm. This light is observed on Earth at 496 nm.

(a)
Calculate the redshift of the light from the galaxy

Use the redshift formula:

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}} = \frac{496 - 486}{486} = \frac{10}{486} = 0.0206$$

(b)
Calculate the speed of the galaxy relative to Earth

Use:

$$z = \frac{v}{c} \Rightarrow v = zc = 0.0206 \times 3.00 \times 10^8 = 6.18 \times 10^6 \text{ m s}^{-1}$$

(c)
Explain how redshift provides evidence that the Universe is expanding

- ✔ Redshift means the light’s wavelength has **increased**, which occurs when the source is moving **away** from the observer.
- ✔ Observing redshift in light from **most distant galaxies** suggests they are **receding** from us.
- ✔ This supports the idea that **space itself is expanding**, and **all galaxies are moving away** from each other — consistent with **Hubble’s Law** and the **Big Bang theory**.

🧠 Revision Tips

- Redshift equation: $z = \frac{\Delta\lambda}{\lambda}$, where $\Delta\lambda = \lambda_{\text{obs}} - \lambda_{\text{emit}}$
- For small redshift values, use $v = zc$ (non-relativistic approximation)
- Positive redshift $\Rightarrow$ object is **moving away**
- More distant galaxies show **larger redshifts** $\Rightarrow$ the Universe is expanding
- In open-ended questions like (c), mention **both observation (redshift)** and **interpretation (expanding space)**